2.2 Characteristics of Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.2 (30-minute plan) · Student sheet

I can…

- find the vertex, axis of symmetry, and y-intercept of a quadratic in standard form and graph it.
- find the minimum or maximum value and the domain and range.
- use intercept form to find the x-intercepts and vertex.

Vocabulary · fill in the missing word

Axis of symmetry — the _____ line $x = h$ that divides a parabola into mirror images

Standard form — $f(x) = ax^2 + bx + c$; the axis is $x = -\frac{b}{2a}$ and the _____ is c

Minimum / maximum value — the _____ of the vertex when $a > 0$ / $a < 0$

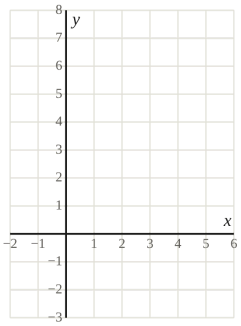
Intercept form — $f(x) = a(x - p)(x - q)$; the _____ are p and q

Part 1 · Notes and examples (fill in each note, then work the example)

Standard form: Standard form: the axis of symmetry is $x =$ _____, and the y-intercept is _____.

Example 1

Graph $f(x) = 2x^2 - 8x + 6$. Label the vertex and the axis of symmetry. Give the minimum value and the range.



Minimum or maximum: $a > 0$: _____ value; $a < 0$: _____ value.

Example 2

Find the minimum or maximum value of $f(x) = -x^2 - 6x + 4$. Give the domain and range.

Intercept form: Intercept form $a(x - p)(x - q)$: the x-intercepts are _____.

Example 3

Find the x-intercepts, the axis of symmetry, and the vertex of $f(x) = -2(x + 3)(x - 1)$.

Part 2 · Partner practice

Partner practice

Find the vertex and the axis of symmetry of $f(x) = x^2 + 4x - 1$.